

Magnetic resonance imaging of mean cell size in human breast tumors

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Purpose: Cell size is a fundamental characteristic of all tissues, and changes in cell size in cancer reflect tumor status and response to treatments, such as apoptosis and cell-cycle arrest. Unfortunately, cell size can currently be obtained only by pathological evaluation of tumor tissue samples obtained invasively. Previous imaging approaches are limited to preclinical MRI scanners or require relatively long acquisition times that are impractical for clinical imaging. There is a need to develop cell-size imaging for clinical applications.

Methods: We propose a clinically feasible IMPULSED (imaging microstructural parameters using limited spectrally edited diffusion) approach that can characterize mean cell sizes in solid tumors. We report the use of a combination of pulse sequences, using different gradient waveforms implemented on clinical MRI scanners and analytical equations based on these waveforms to analyze diffusion-weighted MRI signals and derive specific microstructural parameters such as cell size. We also describe comprehensive validations of this approach using computer simulations, cell experiments in vitro, and animal experiments in vivo and demonstrate applications in preoperative breast cancer patients.

Results: With fast acquisitions (~7 minutes), IMPULSED can provide high-resolution (1.3 mm in-plane) mapping of mean cell size of human tumors in vivo on clinical 3T MRI scanners. All validations suggest that IMPULSED provides accurate and reliable measurements of mean cell size.

Conclusion: The proposed IMPULSED method can assess cell-size variations in tumors of breast cancer patients, which may have the potential to assess early response to neoadjuvant therapy.

KEYWORDS

cell size, density, diameter, diffusion, IMPULSED, MRI, OGSE, oscillating gradient

1 | INTRODUCTION

Cell size is a basic feature of living cells that plays an important role from the molecular to the organ level, including cellular metabolism,¹ proliferation,² and tissue growth.³ Cell size may vary significantly during disease progression or after therapy. For example, cells swell significantly after acute stroke,^{4,5} and cell shrinkage is a hallmark of apoptotic cell death.⁶ For cancer diagnosis and prognosis, cell size is of interest because it varies during mitosis and before death and thus may provide a unique means to evaluate tumor progression and response to treatments. Measurements of cell sizes are reportedly capable of differentiating cancer types⁷ and monitoring tumor early therapeutic response by detecting treatment-induced apoptosis^{8,9} or mitotic arrest.¹⁰ Therefore, quantitative microstructural measurements such as cell size may provide specific means to probe the status of cancerous tissues and would be of potential value in preclinical and clinical applications. Currently, such microstructural information is obtained during conventional clinical care only by invasive biopsies, which are limited not only by the potential to miss important changes attributable to tumor heterogeneity and the small sample size of each specimen, but also may introduce various clinical complications, including pain, hemorrhage, infection, and even death.¹¹ Therefore, a noninvasive imaging technique capable of characterizing tissue microstructural information would be of great interest to clinicians.

Diffusion-weighted MRI (DWI) is an exogenous-agent-free non-invasive imaging technique that provides unique capabilities to probe biological tissue microstructure by evaluating the degree of restriction and hindrance to the free motion of randomly diffusing water molecules. Values of the apparent diffusion coefficient (ADC), a metric obtained using DWI, have been found to be sensitive to cell density and hence are widely used to evaluate cellularity changes after anticancer treatment.¹²⁻¹⁵ However, ADC represents an overall diffusion property of water molecules inside each image voxel, and ADC values are influenced by several tissue parameters simultaneously, including, but not limited to, cell size,¹⁶ cell membrane permeability,¹⁷ intra- and extracellular diffusion coefficients,¹⁸ and intracellular volume fraction.¹⁹ As a result, ADC and tumor cellularity are not always strongly correlated,²⁰⁻²² so that ADC does not reliably provide specific information on cell size and density. Recently, numerous attempts have been made to enhance the specificity of DWI measurements, such as the DDR (diffusion dispersion rate),²³ VERDICT (vascular, extracellular, and restricted diffusion for cytometry in tumors),²⁴⁻²⁶ quantitative temporal diffusion spectroscopy (qTDS) and

IMPULSED (imaging microstructural parameters using limited spectrally edited diffusion),^{27,28} and POMACE (pulsed and oscillating gradient MRI for assessment of cell size and extracellular space) approaches.²⁹ Some of these, for example qTDS and IMPULSED, exploit the dependence of ADC on the time scale (the diffusion time) over which diffusion affects the measured signals, which, in principle, enables the derivation of the spatial scales of restrictions to free displacements.^{27,28} However, previous reports of their applications were either implemented on animal MRI scanners only and used much stronger diffusion gradient amplitudes than are available on clinical MRI scanners, or the total acquisition times were long for clinical applications. Therefore, there is still a need to develop a fast (<10 minutes) and quantitative DWI method that is capable of measuring cell size on clinical MRI machines.

Here, we introduce a modified implementation of IMPULSED for clinical MRI. Instead of using apodized cosine oscillating diffusion gradients as reported earlier,^{27,28} we used IMPULSED acquisitions that incorporate cosine-modulated oscillating trapezoidal diffusion gradients³⁰⁻³² that are readily capable of running on clinical MRI machines, as shown in Figure 1. An IMPULSED protocol incorporates a set of DWI sequences, each of which uses a different diffusion time, with a diffusion time range that makes the DWI images highly sensitive to variations in cell size in human tissues, and from which mapping cell size and density can be derived. In the current work, we demonstrate a practical imaging protocol that combines different DWI sequences, along with new analytical equations that link the DWI signals using the real gradient waveforms to specific microstructural parameters such as cell size. We also provide validations of the IMPULSED method using computer simulations and experimental measurements of cells in vitro and animal models in vivo, along with practical demonstrations of clinical applications in breast cancer patients.

2 | METHODS

2.1 | Pulse sequence

Figure 1 shows the pulse sequences used to acquire DWI data for the IMPULSED method on clinical 3T scanners. In addition to a conventional pulsed gradient spin echo (PGSE) sequence, which measures ADC over longer diffusion times, IMPULSED also uses cosine-modulated, trapezoidal, oscillating gradients in spin echo sequences (OGSE) to measure ADC over different, shorter diffusion times. The combination

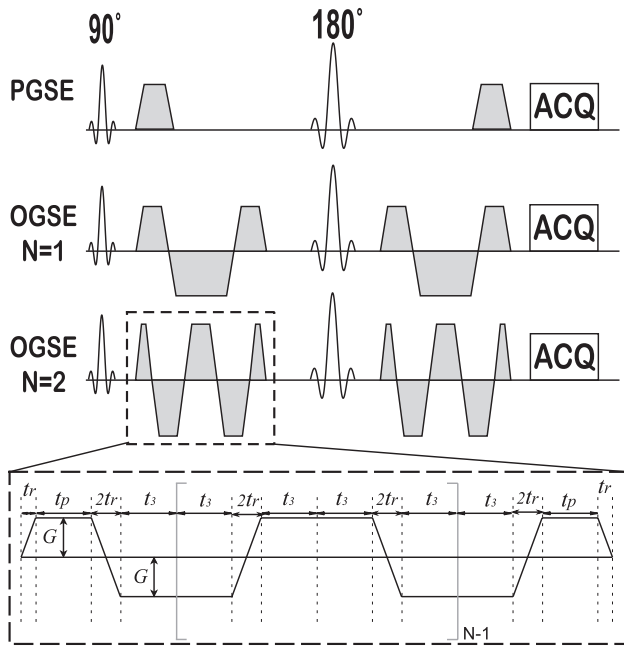


FIGURE 1 Diagram of the pulse sequences used in the IMPULSED method. In addition to conventional PGSE acquisitions, OGSE acquisitions with two frequencies ($N = 1$ and 2) are used. ACQ = acquisition

of longer and shorter diffusion times ensures that ADC values from each sequence will differ, and these differences then reflect the length scales of major restrictions to diffusion, which in tumors correspond to cell sizes.³³ This combination enables detection of a broad range of length scales, providing more-comprehensive information on tissue microstructure than single measurements of ADC.^{27,28} For all diffusion sequences, G is the gradient strength, δ is the duration of each diffusion gradient, Δ is the separation of 2 gradients, t_r is the gradient rise time ($= 0.9$ ms on Philips Achieva 3T MRI scanner; Philips, Best, The Netherlands), t_p is the duration time of the first gradient plateau, and $t_3 = t_p + t_r/2$ for OGSE sequences. N is the number of cycles in each diffusion gradient in the OGSE sequence; $b = \gamma^2 G^2 [(t_r + t_p)^2 (\Delta - (t_r + t_p)/3) + t_r^3/30 - (t_r + t_p)t_r^2/6]$ for the PGSE sequence and $b = \gamma^2 G^2 [91Nt_r^3/15 + 8Nt_p^3/3 + t_r^3/30 + 12Nt_r t_p^2 + 46Nt_r^2 t_p/3]$ for the OGSE sequences^{30,31} shown in Figure 1.

2.2 | Theory

Quantitative information on cell size and density are obtained by fitting a simple model of tissue water to the DWI experimental data. Like previous reports,^{24,27,29,34} DWI signals in IMPULSED are modeled as the sum of signals arising from 2 compartments, that is, intra- and extracellular spaces. The measured DWI signal S is shown by Equation 1:

$$S = v_{in} \cdot S_{in} + (1 - v_{in}) \cdot S_{ex} \quad (1)$$

where S_{in} and S_{ex} are the signal magnitudes per volume from the intra- and extracellular spaces, respectively, and v_{in} is intracellular water fraction. Note that water exchange between intra- and extracellular spaces is ignored, but does not affect the estimation of mean cancer cell size, although it may bias the estimation of cell density.³⁵

2.2.1 | Modeling intracellular diffusion (the first term in Equation 1)

Water molecules are restricted inside cells because cell membranes have only finite permeabilities. For simplification, cancer cells are usually modeled as impermeable spheres, and analytical expressions similar to previous reports³⁴ may be derived to link DWI signals and underlying microstructural parameters such as cell size. The derived analytical equations for cosine-modulated, trapezoidal-shaped OGSE sequences can be found in Supporting Information Equations S5 and S6. The computer simulation validation of the accuracy of PGSE and OGSE equations using real gradient waveforms are shown in Supporting Information Figures S1 and S2.

2.2.2 | Modeling extracellular diffusion (the second term in Equation 1)

It is challenging to obtain an explicit analytical form to describe extracellular diffusion. When the diffusion time range is limited, previous studies have suggested that the extracellular diffusion coefficient shows an approximately linear dependence on gradient frequency (proportional to the inverse of diffusion time).^{23,36} However, partly because of hardware limitations and by the nature of the cell sizes found in real tissues, the frequency range acquired or available on clinical MRI scanners is narrow, so that the dependency of extracellular diffusion on frequency is minor. All our investigations using simulations *in silico*, cell lines *in vitro*, and animals *in vivo* have suggested that the extracellular diffusion coefficient obtained using the IMPULSED method is largely insensitive to diffusion times, so that the extracellular diffusion coefficient is modeled as a constant D_{ex} .

2.2.3 | Diffusion anisotropy

Except for some *ex vivo* investigations of fixed breast tissues with strong gradients,³⁷ a previous *in vivo* study involving 81 patients reported that water diffusion is nearly isotropic in various human breast tumors.³⁸ Therefore, the DWI signals for IMPULSED acquisitions are obtained by

averaging 3 acquisitions with diffusion gradients along 3 orthogonal oblique directions ($[g_x, g_y, g_z] = [1, 1, -0.5], [1, -0.5, 1], \text{ and } [-0.5, 1, 1]$), corresponding to the trace of a diffusion tensor.

2.2.4 | IMPULSED outcomes

Table 1 lists tissue parameters that can be determined by analyses of DWI data. The estimates of d are actually volume-weighted mean cell sizes,³⁹ which, for a population of cells, is given by $d = \sum_n d_n^4 / \sum_n d_n^3$, where d_n is the cell size of the n^{th} cell. Note that, unlike our preclinical animal studies,^{27,28} D_{in} is fixed as $1.58 \mu\text{m}^2/\text{ms}$,¹⁰ which better stabilizes the fittings of IMPULSED data. Supporting Information Figure S4 shows that the fitting precisions of d and v_{in} significantly decrease if D_{in} is a free-fitting parameter. Our simulations suggest that different choices of D_{in} in the data fitting have little influences on the fitted IMPULSED metrics (see Supporting Information Figure S5).

2.2.5 | Experimental diffusion parameters in IMPULSED

It is desirable to use higher gradient strengths and slew rates in diffusion measurements to achieve greater diffusion weighting (b values) and a range of diffusion times appropriate for each sample of interest.⁴⁰ However, because of human physiological thresholds and hardware limitations, the gradient strength and slew rate are limited on clinical MRI scanners, which makes it challenging to implement some quantitative DWI methods. To ensure that IMPULSED can be translated clinically, a maximum gradient strength of 80 mT/m and a slew rate <100 mT/m was assumed, and these limits were imposed on our simulations and experiments including cells, animals, and humans. Table 2 shows diffusion parameters for IMPULSED measurements that are readily available on human 3T MRI systems and were selected for practical implementation. All studies acquired 2 opposite diffusion gradient directions for each axis, and the geometric means were used as final images to mitigate cross-terms between diffusion and background gradients.⁴¹

TABLE 1 Summary of IMPULSED-derived parameters and corresponding biophysical features

IMPULSED parameters	Biophysical features
d	Volume-weighted mean cell size
v_{in}	Intracellular volume fraction
D_{ex}	Extracellular diffusion coefficient

2.3 | Validation using simulations in silico

Computer simulations were performed to evaluate both the accuracy and precision of IMPULSED-derived parameters obtainable for signal-to-noise ratios (SNRs) practically available on clinical 3T MRI scanners. A finite difference method was used to simulate DWI signals as reported previously.⁴² Tumors were modeled as tightly packed spherical cells on a face-centered cubic lattice⁴³ with $v_{in} = 61.8\%$, $D_{in} = 1.58 \mu\text{m}^2/\text{ms}$,¹⁰ $D_{ex} = 2 \mu\text{m}^2/\text{ms}$, and homogeneous relaxation times for simplicity. Eight different values of cell diameter d evenly distributed from 6 to 20 μm were evaluated, covering the cell sizes typical of lymphocytes to cancer cells. After noise-free DWI signals were calculated for each cell diameter, Rician noise equivalent to achieve an SNR = 20 was added, and then the noisy signals were used for data fitting. This process was repeated 100 times to evaluate both accuracy and precision of IMPULSED-derived metrics.

2.4 | Validation using cell lines in vitro

2.4.1 | Cells

Three types of breast cancer cell lines, MDA-MB-231, MCF7, and MDA-MB-453, were purchased from American Type Culture Collection (Manassas, VA). In addition to cancer cell lines, the Jurkat acute T-cell leukemia cell line and lymphocytes were used to mimic smaller cells. Jurkat cells were generously provided by Dr James Thomas at Vanderbilt University. Lymphocytes were extracted from human peripheral blood by using the Ficoll method.⁴⁴ The details of cell preparation can be found in the Supporting Information.

For MR experiments, cells were washed with phosphate-buffered saline; after fixation, approximately 3×10^7 cultured cells (or 1×10^9 lymphocytes) were centrifuged at 2000g for 2 minutes in a 0.65-mL Eppendorf tube to obtain a tight cell pellet. The supernatant was carefully removed for MRI measurements. A small aliquot of cells from each sample was spotted on a glass slide and covered by a coverslip. Digital images of the cells were recorded at both 20 $\times$ and 40 $\times$ amplification. A stage micrometer was used for size calibration.

2.4.2 | MRI experiments

All MRI measurements of cells in vitro were performed on a Varian/Agilent 4.7T MRI spectrometer (Agilent Technologies, Santa Clara, CA) similar to the approach described previously.³⁵ A 2-mm-thick slice crossing the center

TABLE 2 Summary of diffusion parameters used in IMPULSED measurements which are readily available for human 3T MRI systems

	δ/Δ [ms]	N	f [Hz]	b [s/mm ²]	G_{max} [mT/m]
PGSE	12/74	N/A	N/A	0, 250, 500, 750, 1000, 1400, 1800	50.0
OGSE	40.9/51.4	1	25	0, 250, 500, 750, 1000	72.7
		2	50	0, 100, 200, 300	80.5

G_{max} is the maximum gradient strength used with all gradients on.

of each cell pellet was imaged with a field of view (FOV) 16×16 mm and a matrix size 64×64 , yielding a spatial resolution of 250 μ m. All diffusion sequence parameters were the same as in Table 2, and other acquisition parameters are the same as in the animal study in vivo.

2.4.3 | Light microscopy

Bright field images were captured using a Zeiss Axio Observer microscope (Carl Zeiss, Jena, Germany). Two representative FOVs were chosen for each slide. Each FOV was imaged at 40 $\times$ magnification with the focus set slightly above, below, and equal to the optimal focal plane. Differences between 2 out-of-focus images resulted in enhanced contrast of the cell boundaries.⁴⁵ The area of each cell was then calculated from these microscopic images using an autosegmentation program written in Matlab (The MathWorks, Inc.), and these measurements were converted to a diameter assuming each cell is a sphere. The detailed procedures are provided in the Supporting Information, and representative raw and segmented light microscopy images are shown in Supporting Information Figure S6.

2.5 | Validation using animals in vivo

2.5.1 | Animal preparation

All procedures were approved by the Vanderbilt University Institutional Animal Care and Usage Committee. MDA-MB-231 ($n = 6$) and MCF7 ($n = 4$) xenografts were generated following subcutaneous injection of 1 to 2×10^6 cells in female athymic nude mice (Harlan Laboratories, Inc., Indianapolis, IN). When each tumor reached a size of 200 to 300 mm³, MRI was performed as described below, and each mouse was euthanized for histology immediately afterward.

2.5.2 | MRI experiments

All MR images of animals were acquired on a 4.7T Varian/Agilent horizontal small animal scanner (Agilent

Technologies) using a Litz38 volume coil for both transmission and reception. A single-shot echo-planar imaging (EPI) diffusion sequence with fat suppression was used for all diffusion measurements to minimize motion artefacts. Axial slices with 1-mm thickness were acquired to cover the entire tumors, with in-plane matrix size 128×64 and FOV = 40×20 mm yielding an in-plane resolution of 312.5×312.5 μ m. All diffusion sequence parameters are outlined in Table 2: TR/TE = 4500/104 ms; receiver spectral width 250 kHz; and half scan factor = 0.875.

2.5.3 | Histology

Animals were euthanized immediately following MRI scans. Tumors were collected, cut into 2-mm-thick pieces, and fixed in 10% formalin for 24 hours. Tissues were transferred to 70% ethanol before paraffin embedding. Tumors were sectioned at 8- μ m thickness and stained with hematoxylin and eosin (H&E) for cellularity and with Na⁺/K⁺-ATPase (ab76020; Abcam, Cambridge, MA) for cell size.

2.6 | Applications in patients

2.6.1 | Breast cancer patients

The human imaging study was approved by the institutional review board at Vanderbilt University Medical Center. Seven women (age, 55.3 ± 8.0) diagnosed with breast cancer with tumors ≥ 1 cm were recruited in this study. Written informed consents were received from participants before inclusion in the study. Detailed patient information and tumor types can be found in Supporting Information Table S1.

2.6.2 | Human MR imaging

IMPULSED imaging was performed using a Philips Achieva 3T scanner with a 16-channel breast coil. Acquisition sequence parameters were: TR/TE = 4500/103ms; FOV = 192×192 mm; acquisition matrix size, 96×96 ; reconstructed

in-plane resolution = 1.3×1.3 mm; 10 or 20 slices; slice thickness = 5 mm; single-shot EPI; sensitivity encoding factor = 3; half scan factor 0.64; water-fat shift (pixels)/bandwidth (Hz) = 12.181/35.7; fat suppression with spectral adiabatic inversion recovery; and dynamic stabilization was used to minimize DWI signal drifts during scanning. All diffusion sequence parameters were the same as in Table 2. Total scan time was ≈ 7 minutes. In addition, ADC measurements using PGSE acquisitions with $\Delta = 54$ and 34 ms were performed to further investigate ADC dependence on diffusion times.

2.7 | Data analyses

All diffusion images were coregistered to the corresponding T_2 -weighted $S(b = 0)$ image to correct for subject motion. For cell experiments in vitro, a region of interest (ROI) was manually drawn on each image, and the total DWI signals from the ROI were used in the data fitting. Light microscopy images of cells were analyzed using a locally developed pipeline, with details provided in the Supporting Information. For in vivo experiments, tumor ROIs were manually drawn based on PGSE diffusion-weighted images with $b = 1000$ s/mm². All ROIs were drawn without boundary voxels to avoid partial volume effects. IMPULSED fitting was performed only inside ROIs. Histology images were analyzed using CellProfiler, and the details are provided in the Supporting Information.

All data fittings were performed using Matlab (The MathWorks, Inc.) to generate DWI parametric maps on a voxel-wise basis.³⁹ The fitting parameter ranges were limited by possible physiologically relevant values, that is, $0.2 \leq d \leq 25$ μm (i.e., typical breast cancer cell size range; note that the upper limit is determined by the root mean square displacement of free water diffusion at 37°C),

$0 \leq v_{in} \leq 1$ (maximum volume fraction, 100%), and $0 \leq D_{ex} \leq 3.1$ $\mu\text{m}^2/\text{ms}$ (the free water diffusion coefficient is 3.07 $\mu\text{m}^2/\text{ms}$ at 37°C). D_{in} was fixed as 1.58 $\mu\text{m}^2/\text{ms}$ ¹⁰ in the data analysis to stabilize fittings as shown in Supporting Information Figure S5. Note that fitting results are insensitive to the choices of D_{in} . Fittings were to maximize the log likelihood function with Rician noise, that is, $L_R = \sum_{n=1}^N \left[\log M_n - 2 \log \sigma + \log I_0 \left(\frac{S_n M_n}{\sigma^2} \right) - \frac{S_n^2 + M_n^2}{2\sigma^2} \right]$, where S_n and M_n are the model-predicted and measured signals of the n^{th} measurements, respectively, I_0 is the modified Bessel function of the first kind with order 0, and σ^2 is the noise variance. In the human study, all diffusion images and IMPULSED-derived parametric images were coregistered to corresponding high-resolution T1-weighted anatomical images using the FMRIB's Linear Image Registration Tool (FLIRT)⁴⁶ in the FSL toolbox.⁴⁷

Spearman's correlation was calculated to determine the relationship between fitted cell-size and histology values at the ROI level. The Bland-Altman plot was used to evaluate the agreement of cell-size values between those from fitted and those derived from histology.

3 | RESULTS

3.1 | Computer simulations in silico

Figure 2 shows the simulated dependence of IMPULSED-derived metrics on input values with an SNR of 20, typical of human MRI. In the range 6 to 20 μm , the fitted cell size d shows a clear linear dependence on ground-truth values. A mixed linear model reports $P < 0.01$. The fitted intracellular water fraction v_{in} shows a good match to any ground-truth value with a Bonferroni-adjusted $P < 0.05$. By contrast, the fitted extracellular diffusion coefficient

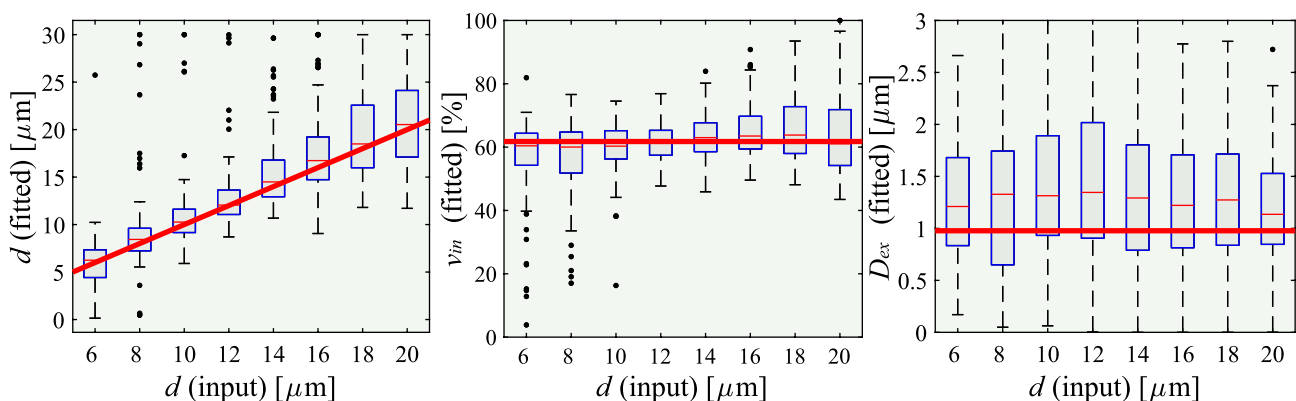


FIGURE 2 Simulated influence of noise with SNR = 20 on fitted d (left), v_{in} (middle), and D_{ex} (right) using IMPULSED. For each real input d , the fittings were repeated 100 times each with different noise samples, but with the same SNR level. The red solid lines represent the ground-truth values, boxes represent ranges between the 25th and 75th percentiles, and dots are outliers

D_{ex} shows significant uncertainties, indicating that the precision of fitted D_{ex} is limited when SNR is low. Note that the fitted D_{ex} is expected to be lower than the intrinsic extracellular diffusion coefficient because of restriction effects. If SNR increases to 50, the precisions of fitted d , v_{in} , and D_{ex} can be dramatically improved (see Supporting Information Figure S3). In conclusion, although limited in practice by the available gradient strength and slew rate,

IMPULSED can measure mean cancer cell size reliably when $6 < d < 20 \mu\text{m}$ with practical SNRs, but the estimation of extracellular diffusion coefficient is not reliable with low SNRs.

3.2 | Imaging cells in vitro

Supporting Information Figure S6 shows an example of the cell segmentation of light microscopy images. The cell-size information obtained from microscopy was assumed to be the ground truth for validating the cell sizes fitted using IMPULSED. Figure 3 compares the mean cell sizes obtained from IMPULSED fitting and light microscopy for 3 breast cancer cell lines, the Jarkat cell line, and lymphocytes. Over a broad range of cell sizes (11–18 μm) and cell types (breast cancer, leukemia, and lymphocytes), IMPULSED fitted d values show a strong correlation ($r = 0.92$ and $P < 0.001$, provided by the Pearson correlation) with the values obtained from light microscopy.

3.3 | Imaging animals in vivo

Figure 4 shows representative multiparametric images of a mouse MDA-MB-231 xenograft. The T_2 -weighted $S(b=0)$, diffusion-weighted OGSE with $N=1$ and $b=1000 \mu\text{m}^2/\text{ms}$ and PGSE with $b=1000 \mu\text{m}^2/\text{ms}$ images are shown in the top

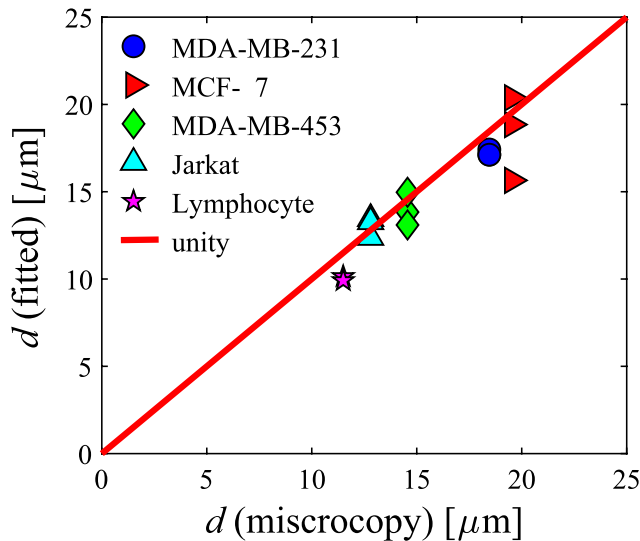


FIGURE 3 Comparison of mean cell sizes obtained from IMPULSED fitting and light microscopy

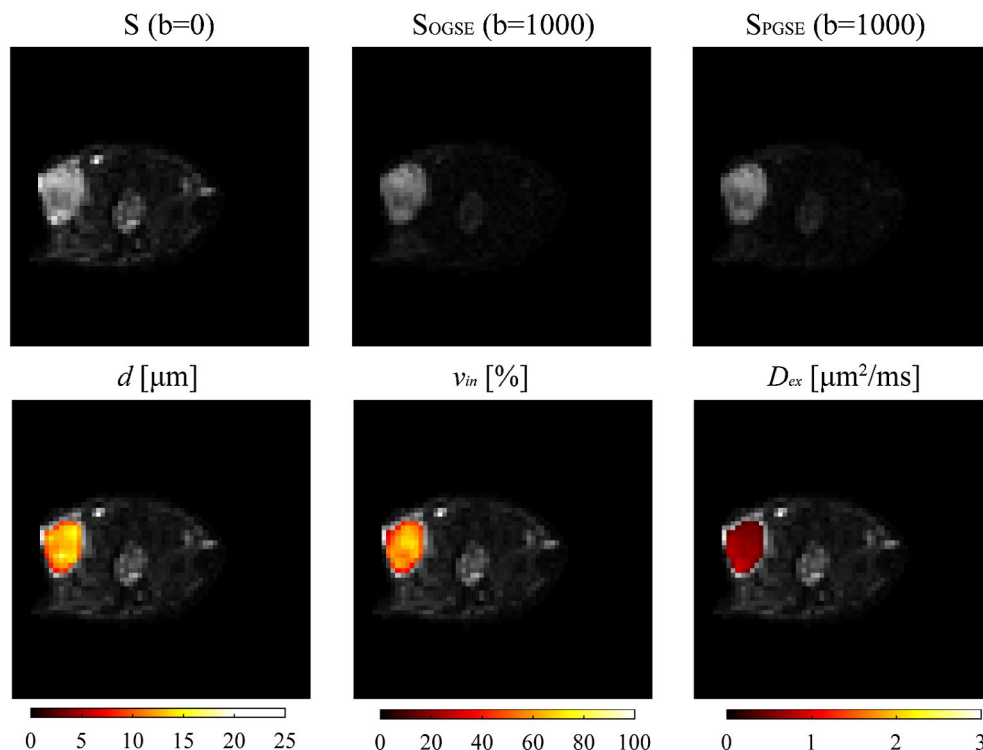


FIGURE 4 Representative multiparametric images of a mouse bearing MDA-MB-231 xenograft. (Top) T_2 -weighted $S(b=0)$, OGSE with $N=1$ and $b=1000 \mu\text{m}^2/\text{ms}$, and PGSE with $b=1000 \mu\text{m}^2/\text{ms}$. (Bottom) IMPULSED-derived maps of mean cell size d , intracellular volume fraction v_{in} , and extracellular diffusion coefficient D_{ex} overlaid on the corresponding S_0 image

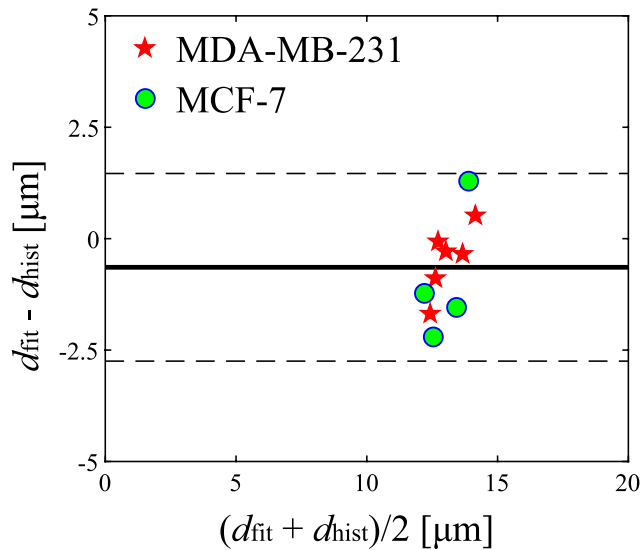


FIGURE 5 Bland-Altman plot of fitted and histologically obtained mean cell sizes of 2 types of breast cancer xenografts. The solid line represents the mean difference, and 2 dashed lines represent 95% of limits of agreement (i.e., 1.96 SD). SD = standard deviation

row. IMPULSED-derived maps of d , v_{in} , and D_{ex} were overlaid on the corresponding diffusion-weighted image shown in the bottom row. There is significant inhomogeneity within the tumor. Figure 4 Representative multiparametric images of a mouse bearing MDA-MB-231 xenograft. (Top) T_2 -weighted $S(b=0)$, OGSE with $N=1$ and $b=1000 \mu\text{m}^2/\text{ms}$, and PGSE with $b=1000 \mu\text{m}^2/\text{ms}$. (Bottom) IMPULSED-derived maps of mean cell size d , intracellular volume fraction v_{in} , and extracellular diffusion coefficient D_{ex} overlaid on the corresponding S_0 image

Figure 5 is a Bland-Altman plot of fitted and histology-derived mean cell sizes for 2 types of breast cancer xenografts, MDA-MB-231 ($n=6$) and MCF7 ($n=4$), imaged in mice in vivo. IMPULSED slightly underestimates d (mean difference = $0.6 \mu\text{m}$), and 95% limits of agreement is $2.1 \mu\text{m}$. This suggests good agreement between the IMPULSED- and histology-derived mean cell sizes in animals in vivo.

3.4 | Imaging breast cancer patients in vivo

Supporting Information Table S1 summarizes the information on breast cancer patients and breast tumors. The SNR was ~ 45 in the tumors. Figure 6 shows IMPULSED signals (symbols) from a tumor ROI and the IMPULSED fits (solid lines) of a representative human breast tumor. The fitted overall average cell size was $d = 14.88 \pm 4.39 \mu\text{m}$, $v_{in} = 38.87 \pm 7.95\%$, and $D_{ex} = 1.81 \pm 0.46 \mu\text{m}^2/\text{ms}$, yielding a cell density = $2.86 \pm 1.31 \times 10^8 \text{ cells}/\text{cm}^3$, which is consistent with previous reports of $\sim 10^8 \text{ cells}/\text{cm}^3$ for tumors of

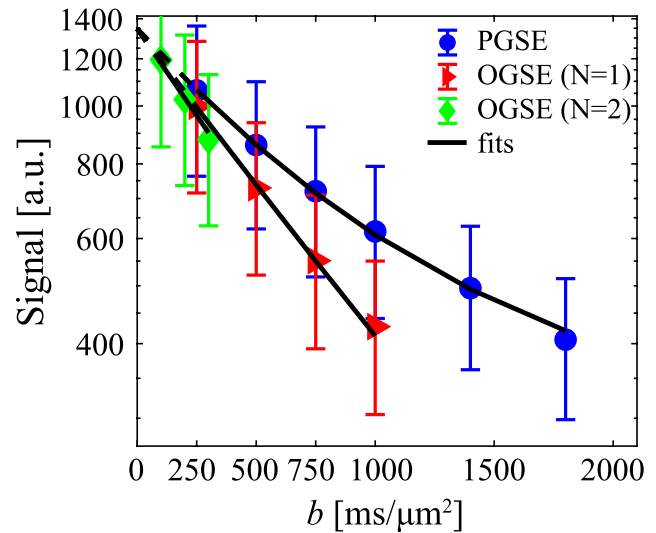


FIGURE 6 The ROI-based diffusion-weighted signal attenuations of a representative human breast tumor. Markers are mean signals, and the error bars represent standard deviations. The solid lines are fitted results using Equation 1, and dashed lines with low b values indicate $b=0$ images were excluded from fittings to minimize perfusion effects

epithelial origin (such as breast tumors).⁴⁸ Note that for the same b values, signals acquired with OGSE with an effective diffusion time t_{diff} of 10 ms decay significantly more than those obtained with PGSE with $t_{diff} = 70 \text{ ms}$. This increase in ADC at shorter diffusion times obtained provides the contrast that enables the possibility to measure cancer cell size and density. An example of ADC dependence on diffusion times can be found in Supporting Information Figure S8. For the IMPULSED fitting, all $b=0$ images were excluded from the fittings in order to minimize the influences of blood perfusion.

Figure 7 shows multiparametric images of all 7 patients. In addition to the large variations of tumor sizes across all patients, the parametric maps are significantly inhomogeneous within each tumor. For example, there are significantly lower intracellular volume fraction and increased extracellular diffusion coefficient at the center of the tumor of patient 2, suggesting that an acellular, necrotic core has developed. Note that the parametric map of D_{ex} is more heterogeneous inside tumors, which may be attributable to the lower fitting precision (see Figure 1).

Figure 8 summarizes the histograms of all fitted IMPULSED metrics for 7 patients. Although different patients and breast tumors show different histograms of the IMPULSED metrics, the peaks of d are all in the range of 12 to $18 \mu\text{m}$ and v_{in} are in the range of 25% to 40%. Values of v_{in} may be underestimated because of transcytolemmal water change, but this does not affect d .³⁵ D_{ex} shows broader ranges of distribution compared with those of d and v_{in} . This may be attributable to their larger fitting variations as predicted

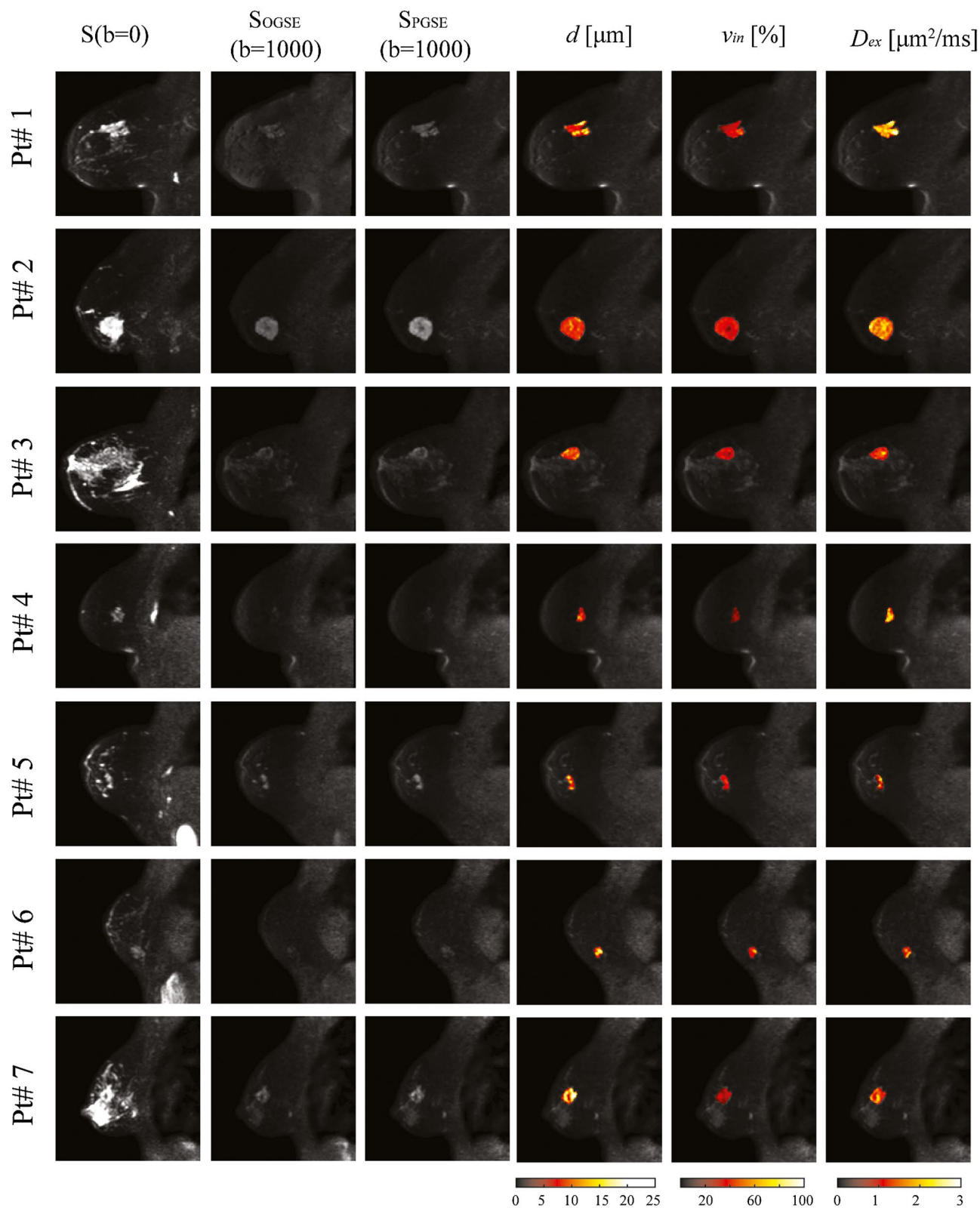
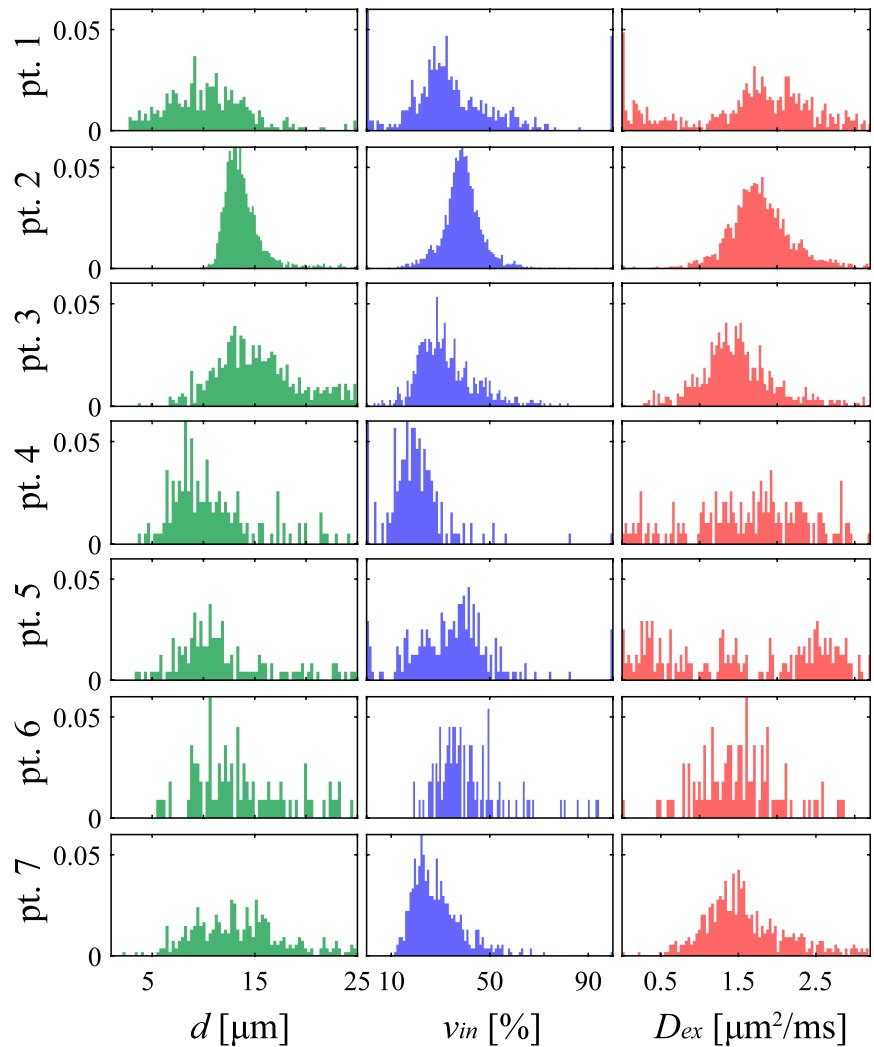


FIGURE 7 Representative multiparametric images of all 7 patients. (From left to right) T_2 -weighted $S(b=0)$, OGSE with $N=1$ and $b=1000 \mu\text{m}^2/\text{ms}$, PGSE with $b=1400 \mu\text{m}^2/\text{ms}$, IMPULSED-derived maps of mean cell size d , intracellular volume fraction v_{in} , and extracellular diffusion coefficient D_{ex} overlaid on the corresponding diffusion-weighted image

FIGURE 8 A summary of histograms of all fitted IMPULSED metrics (columns) of all 7 patients (rows)



by the simulation results (Figure 1). Note that IMPULSED-derived parameters do not correlate with one another (data not shown), which is not surprising given that they represent independent tissue features.

4 | DISCUSSION

We have implemented a clinically feasible noninvasive imaging method termed IMPULSED to measure mean cancer cell size of tumors using clinical MRI scanners *in vivo*. The potential clinical relevance of the IMPULSED method is based on the important role of cell size in cancer. For example, 1 potential clinical application is to use cell-size variations to monitor tumor early response to anticancer treatment. As shown in our previous work using animal models, cell-size variation is an early and specific biomarker to report the effectiveness of anticancer treatment, such as in treatment-induced mitotic arrest¹⁰ and apoptosis.⁹ The current work provides a reliable technique that allows measurements of mean cell sizes of human breast tumors in a clinical setting, which

is an essential step to translate our previous preclinical findings to clinical practice. Moreover, it would be interesting to correlate IMPULSED-derived parameters with other pathohistological and/or genetic features of tumors to assess tumor status, perhaps as part of a “radiomics” characterization of cancers. However, this is out of the scope of this article.

Despite numerous attempts at using diffusion MRI to map microstructural information in tumors *in vivo*, successful applications to date have so far been very limited. Although providing similar information such as cell size, IMPULSED and other diffusion-based methods, such as VERDICT and POMACE, use different acquisitions and methods of analysis. It is therefore of interest to compare the accuracy and precision of different methods. The VERDICT approach attempts to map mean cell size in human tumors, but to date has been exclusively used in human prostate cancers.²⁴⁻²⁶ The VERDICT approach is different from IMPULSED in that though VERDICT also uses multiple diffusion times; it uses conventional PGSE sequences with higher *b* values. Consequently, the range of diffusion times probed is narrower than for IMPULSED. Moreover, VERDICT uses multiple

acquisitions with different echo times, which may introduce biases in fitting from relaxation time effects. IMPULSED incorporates cosine-modulated trapezoidal oscillating gradients to significantly increase the range of effective diffusion times in the short time range. Supporting Information Figure S8 shows the diffusion time dependence of ADC of a human breast tumor. Over the range t_{diff} 30 to 70 ms achievable using PGSE sequences and realistic gradient strengths only (such as those used in VERDICT), the ADC does not change significantly. The most significant ADC change occurs between 30 and 5 ms, the range used in IMPULSED. This provides good sensitivity to cellular properties, which forms the foundation of IMPULSED in measuring cell size.

Because this is the first demonstration of IMPULSED in human breast cancer imaging, the experimental parameters were not all optimized, and the protocol was chosen to include more acquisitions to maximize the accuracy. For example, the current IMPULSED method acquires both positive and negative gradients in each direction to remove cross-terms, acquires 3 orthogonal gradient directions to account for any small diffusion anisotropy in breast tumors, and uses a much longer TR (4.5 seconds) to reduce any possible influences from the relatively long T_1 relaxation times in tumors. All these strategies are theoretically valid for any diffusion method. However, their exact influence on IMPULSED-derived parameters remains unclear. Moreover, an optimization of experimental parameters could maintain the precision while minimizing scan time, such as utilized in our previous work using the Cramér-Rao lower bound approach for optimizing T_1 and quantitative magnetization transfer protocols.⁴⁹ Recent optimization efforts have shortened VERDICT acquisition time from 35 minutes²⁵ to 12 minutes,²⁶ suggesting that there is much room to accelerate the acquisition protocol of diffusion MRI-based methods.

Although IMPULSED detects the mean cell size within each image voxel, it is plausible to obtain the distribution of cell sizes inside each voxel as well. However, it is challenging to fit cell-size distributions without a-priori knowledge of their nature.⁵⁰ Most quantitative DWI studies to date have modeled cancer cells as spheres^{10,27} or neural axons as cylinders³⁹ with uniform cell sizes to simplify the mathematical complexity yet preserve basic microstructural features. Although a mean cell size and density can be measured inside each imaging voxel, the voxel-by-voxel heterogeneous spatial distributions of cell size and cell density across whole tumors can still be obtained, as shown in Figure 8.

Not only mean cell size d , but also the intracellular volume fraction v_{in} can be derived from IMPULSED. The apparent cell density can be calculated from d and v_{in} .²⁸ However, our previous study³⁵ found that, if the transcytolemmal water exchange cannot be ignored, the intracellular volume fraction is intrinsically underestimated for any biophysical diffusion model that assumes no water exchange (such as IMPULSED

and VERDICT). Because cell-membrane permeability is likely to increase during treatment-induced apoptosis, the accuracy of cell density derived from IMPULSED may be compromised. Therefore, caution should be expressed in the interpretation of fitted v_{in} values as intracellular volume fraction. Interestingly, v_{in} has been found to correlate well with ground-truth cell density in simulations,²⁷ which suggests that v_{in} might still be a good indicator of cell density, although its absolute accuracy is uncertain.

Although only breast cancer was investigated in the current study, IMPULSED can be used to assess other extracranial tumors, such as head and neck tumors, wherever a 2-compartment model is valid. Note that IMPULSED cannot specifically differentiate cancer cells from other cells. Therefore, the mean cell size obtained using IMPULSED includes all cell types (e.g., cancer cells, stromal cells, and lymphocytes). Our in vitro cell studies (Figure 3) suggest that IMPULSED has sufficient sensitivity to differentiate small lymphocytes, relatively small cancer cells (leukemia Jarkat), and relatively large breast cancer cells. The relative fraction of these cells will change the mean cell size. If more-detailed information on specific types of cells is needed, the 2-compartment model may need to be modified.

To ensure clinical translation, the IMPULSED method presented in the current study was strictly limited to diffusion parameters that are achievable on clinical 3T MRI scanners, such as a maximum gradient strength <80 mT/m and maximum gradient slew rate <100 mT/m/sec. However, the ability of IMPULSED is not limited by these standard parameters. For example, more-advanced MRI hardware, such as the Human Connectome gradient coil with a maximum gradient strength 300 mT/m and maximum slew rate 200 mT/m/sec, can remarkably improve the ability of diffusion MRI to probe brain microstructure.⁴⁰ Different from fixing intracellular diffusivity D_{in} in the fittings in the current work, we previously found that D_{in} can be fit reliably with a gradient strength up to 360 mT/m using the IMPULSED method.³⁵ Because D_{in} is an important indicator of overall intracellular microenvironment, high-performance gradient systems can remarkably enhance the ability of IMPULSED to characterize intracellular space in human imaging.

5 | CONCLUSIONS

A clinically feasible IMPULSED imaging method has been successfully developed and validated on clinical MRI scanners for in vivo imaging of mean cell sizes of solid tumors in breast cancer patients. To the best of our knowledge, this is the first clinical study that uses a noninvasive imaging method for spatially mapping distributions of mean cancer cell size of heterogeneous human breast tumors in vivo.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

FIGURE S1 Comparison of intracellular PGSE DWI signals obtained from simulation vs analytical forms with trapezoid waveform (Equation S3) and square waveform (Equation S4)

FIGURE S2 Comparison of intracellular DWI signals obtained from simulation vs analytical expressions with trapezoidal OGSE sequences and $N = 1$ (Equation S5) and $N = 2$ (Equation S6)

FIGURE S3 Simulated influence of noise with $\text{SNR} = 50$ on IMPULSED-derived metrics. For each real input d , the fittings were repeated 100 times each with different noises, but with the same SNR level. The red solid lines represent the ground-truth values inputted in the simulations, boxes represent ranges between the 25th and 75th percentiles, and dots are outliers

FIGURE S4 Simulated IMPULSED-derived metrics with $\text{SNR} = 20$ if D_{in} is a free-fitting parameter

FIGURE S5 Simulated influence of noise with $\text{SNR} = 20$ on IMPULSED-derived metrics with different choices of fixed D_{in} values in the data analyses. From the top to bottom, fixed $D_{in} = 1.0, 1.4, 1.58$ (ground truth), 1.8, and $2.2 \mu\text{m}^2/\text{ms}$

FIGURE S6 A representative FOV of light microscopy of cells in vitro. The left is the raw image, and the right shows the segmented images with identified cells highlighted in green

FIGURE S7 A representative image of an MDA-MB-231 breast tumor in mouse. The top is the raw image, and the bottom is the images after the processing of CellProfiler

FIGURE S8 Tumor ADC dependence on diffusion time. The error bars represent the standard deviations over the tumor region shown in Figure 4

Table S1 Information on breast cancer patients and their tumor types

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